MODULE-WISE QUESTION BANK

Academic Year :	2026-27	**Batch :**	2024
Course Code :	BCS502	**Course Title :**	Computer Network and Security
Credits :	4	**Course Coordinator :**	Prof. Nagamani D R
Total Contact Hours :	**40L+20P**	**Total Hours of Pedagogy:**	40+20
Faculty In-charge :	Dr. Jayasheela C S, Prof. Deeksha C, Prof. Roopa B	**Semester & Section**	5th (A, B, & C)

MODULE - 2:Network Layer: Network Layer Services, Packet Switching, Network Layer Performance, IPV4 Addresses and Routing algorithms

Q. No.	Question	CO	RBL
1	Explain the services provided by the network layer in the OSI model.	1	L2
2	What is packet switching? Explain the virtual-circuit approaches to packet switching.	1	L2
3	Explain the performance parameters of the network layer, including delay, throughput, and packet loss.	1	L2
4	Explain IPv4 addressing and describe the different classes of IPv4 addresses.	1	L2
5	Explain Class A, B, C, D, and E IPv4 address classes. What are the disadvantages of classful addressing?	1	L2
6	Explain CIDR (Classless Inter-Domain Routing) with an example. How does CIDR reduce IP address wastage?	1	L2
7	Differentiate between classful and classless addressing. Explain CIDR with an example.	1	L2
8	Explain the format of an IPv4 datagram and the functions of its major header fields.	1	L2
9	Explain fragmentation and reassembly in IPv4. Why are they required?	1	L3
10	Explain the distance-vector routing algorithm with a suitable example.	1	L3
11	the link-state routing algorithm and describe its major steps.	1	L3
12	Explain the structure of the Internet, including Internet Service Providers (ISPs) and their interconnections.	1	L3
13	An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.	2	L3
14	A college has been assigned the IPv4 address 192.168.10.0. The network administrator wants to connect 200 computers in the computer laboratories. Answer the following questions: a) Identify the class of the given IP address under classful addressing.	2	L3

	b) Determine the default subnet mask. c) Is the assigned network sufficient to accommodate 200 computers? Justify your answer. d) Suggest a suitable CIDR prefix and determine the number of usable host addresses.		
15	A university network consists of five routers (A, B, C, D, and E). The link costs are as follows: • A–B: 2 • A–C: 5 • B–C: 1 • B–D: 4 • C–D: 2 • C–E: 6 • D–E: 1 Router A must send a packet to Router E. Answer the following questions: a) Identify the shortest path from A to E using Dijkstra's algorithm. b) Calculate the total cost of the shortest path. c) Explain how Dijkstra's algorithm determines the shortest path. d) If the cost of link D–E increases to 10, determine the new shortest path and its cost.	2	L3

Faculty In-charge

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**